

Sponsorship and Recall of Sponsors

T. Lardinoit

E.S.S.E.C.

C. Derbaix

F.U.C.A.M.

ABSTRACT

The effect of field and/or television sponsorship on respondent's unaided recall and aided recall (recognition) of sponsors' names was explored by means of a laboratory experiment. Findings are discussed with reference to respondents' involvement with a sport.
© 2001 John Wiley & Sons, Inc.

Marketers increasingly believe that sponsorship provides an alternative to traditional advertising. The value of sponsorship, expressed as a percentage of worldwide advertising expenditure, has risen from 2.5 to 3.5% in 1986 to 5.7% in 1996 (Meenaghan, 1998). One of the reasons for this increase is that sponsorship can contribute to reaching unprecedented audiences, both in terms of target diversity and quantity, as shown by the 37 billion television viewers of the 1998 Soccer World Cup.

Accordingly, sponsorship has become an increasingly heavy investment for marketers. First, the property rights to individual sponsorships are more costly (e.g., an official partner of FIFA during the World Cup paid only \$15 million in 1994, paid \$27 million in 1998, and will pay \$30 million in 2002) (Crimmins & Horn, 1996; "Le Nouvel Economiste" 1998). Second, marketers may occasionally spend up to 10 times these amounts on leveraging their initial investments in sponsorship rights (Meenaghan, 1994, 1998; Pham & Johar, 1997).

In the face of the rising costs of sponsorship and support promotional

spending, marketers are beginning to question the cost effectiveness of sponsorship. Therefore, evaluation of the impact of different types of sponsorship is a key issue for marketing managers and consumer researchers.

The financial implications are important, because, for instance, television and field sponsorships come at very different prices. Knowing that some consumer brand-choice decisions are not dependent on unaided recall (Krugman, 1972, 1985) and that reaching a given level of recognition demands less communication investment than reaching a similar level of recall (Singh, Rothschild, & Churchill, 1988), the question then arises as to whether it is appropriate to invest in a single form of sponsorship (e.g., television sponsorship or field sponsorship), or whether a combination of forms is necessary.

Recent major events show the relevance of studying the particular combination of field and television sponsorship currently in widespread use in global events (e.g., World Cup Soccer, Tennis ATP tour, Formula One Grand Prix, Le Tour de France, and World Track and Field Championship). Some marketers have begun to question the effectiveness of sponsorship investment and have recently declared that they would rather not spend the entry fee required by major international sports events (Townley, Harrington, & Couchman, 1998), and instead might allocate their communication budgets to other, supposedly more controllable communication tools with a proven efficiency. Marketing managers need to know whether specific objectives can be achieved by different modes of sponsorship (e.g., TV or field sponsorship) or whether the least expensive sponsorship option will suffice.

This article reports on an experimental test of the effectiveness of different types of sport sponsorship—field, television, and the combination of the two—on television viewers' aided and unaided recall of sponsors' names.

DEFINITION OF SPONSORSHIP

There is a broad consensus regarding the working definition of sponsorship as involving an "investment . . . in an activity . . . in return for access to exploitable commercial potential associated with this activity" (Meenaghan, 1991, p. 36). Building on this understanding, the present article defines sponsorship as ". . . a technique which consists, for any organization, in directly creating or supporting an event that is independent of that organization, and associating itself with that event through the media in order to reach some marketing communication objectives" (Derbaix, Gérard, & Lardinoit, 1994, p. 60).

Field sponsorship refers to the placement of sponsored messages, usually involving a name or logo on sports equipment and billboards located around the event arena. Television sponsorship refers to a system

whereby an advertiser associates its name or trade name with a program or its promotion, excluding any direct or indirect commercial promotion of its products or services (European Group of Television Advertising, 1993). Television sponsorship announcements are shown outside traditional commercial breaks (i.e., during the opening and end credits) of broadcasts of televised sports events or with the promotion of such a program (promotional bumpers). This study focuses on the differing impacts of these two modalities, namely, field and television sponsorship, on the memory of television viewers in terms of recognition and recall.

IMPACT ON MEMORY: FIELD VS. TELEVISION SPONSORSHIP

Field Sponsorship

Field sponsorship stimuli are “messages that are limited to the brand name or to a few words summarizing the brand’s positioning platform” (Pham & Vanhuele, 1997, p. 407). Field sponsorship stimuli by definition do not interrupt either the sponsored event or the televised relay of that event. The centrality of the sports activity being viewed contributes to the low-involvement nature of this kind of message. d’Ydewalle, Vanden Abeele, Van Rensberger, and Coucke (1988) demonstrated that when watching soccer games, television viewers pay attention to sponsor’s billboards <3% of the time. Furthermore, exposure to field sponsorship stimuli does not provide viewers with an opportunity to engage in extensive processing of the message. The level of distraction is very high (Tavassoli, Schultz, & Fitzsimons, 1995), with television viewers essentially devoting their attention to watching the game rather than to processing the field sponsor’s message. Additionally, the audience is faced with a large number of brand names displayed by field sponsors. Thus, the resulting learning process can be limited and incidental.

Television Sponsorship

In the case of broadcast sponsorship of a televised sports event, the tops-and-tails format (i.e., opening and closing credits) means that distractions from the sponsor stimuli caused by the event itself are, by definition, absent. In this format, the viewer’s attention is focused for a few seconds on the television sponsorship stimuli only, giving viewers more opportunity to process the stimuli and allowing better learning. Because a high degree of learning allows better retrieval and thus, better brand-name recall (Pieters & Bijmolt, 1997), it follows that memorization through television sponsorship will outperform memorization through field sponsorship.

Furthermore, because television sponsors’ messages are shown im-

mediately before or after the sports event itself (or in any case outside of regular commercial breaks), primacy and recency phenomena could positively impact on memorization of these stimuli.

Television sponsorship stimuli also involve several modality features (auditory and visual). Keller (1991) suggests it is more beneficial for learning if an item of information is presented in two modalities rather than if it is identically repeated.

For all these reasons, it is hypothesised that:

H1a: The impact of television sponsorship on unaided recall (of the sponsor's name) is greater than that of field sponsorship.

H1b: The impact of television sponsorship on aided recall (of the sponsor's name) is greater than that of field sponsorship.

THE EFFECT OF ENDURING INVOLVEMENT ON THE EFFICIENCY OF FIELD AND TELEVISION SPONSORSHIP

This section seeks to examine the concept of enduring involvement with a sport on the part of a viewer, and the impact this can potentially have on both recognition and recall of sponsor's stimuli, presented singly or in combination as television messages or field displayed messages.

Enduring Involvement

There is an extensive literature (e.g., Celsi & Olson, 1988; Laurent & Kapferer, 1985), which has established that involvement implies—in some cases—an enduring dimension.

Enduring involvement corresponds to a kind of genuine enthusiasm, a strong and solid interest that comes from the relevance of an object or subject for the individual. In the case of sports, it leads the individual to watch events on television, and consequently, it increases the exposure of the viewer to the sponsor's stimuli.

In terms of motivation, a subject with enduring involvement will look for any information that will increase his/her expertise (Richins & Bloch, 1986). Motivation for gathering and processing information leads in the long run to a better knowledge of the domain (Zaichkowsky, 1985). This was confirmed for various sports by a number of researchers, namely, Celsi and Olson (1988, tennis), d'Ydewalle et al. (1988, soccer), and Pham (1992, soccer).

In terms of processing capacity, the more structured the knowledge of a domain, the more able the individual is to activate, from his/her memory, the interpretation of new related information. The greater the number of possible links and associations between new and existing information, the more effective the performance of memory (Okechuku, 1992).

In summary, the characteristics of involved subjects include both their motivation and their capacity to process the information, yielding specific responses to sponsors' stimuli.

Enduring Involvement: The Case of Field Sponsorship

Exposure to field-sponsorship stimuli could in theory have the effect of embedding new memory traces for the individual. Because of the absence of centrality and because of the peripheral nature of the processing in the case of field-sponsorship stimuli, it is likely that such embedded memory traces can only be superficial. A memory trace can arise from such incidental learning, but only where the memory task involves mere discrimination (i.e., recognition). This kind of superficial trace would not normally be evident in unaided recall testing.

Unaided recall is usually understood as a two-stage process requiring both retrieval and discrimination tasks and it depends on both the availability and accessibility of information. Recognition (aided recall), on the other hand, depends merely on the availability of information in memory (Park & Hastak, 1994). Consequently, it is well established that recognition responds better to learning than does unaided recall (Singh, Rothschild, & Churchill, 1988). The impact of field sponsorship is thus too superficial to be significant in terms of deep memorization (which requires both discrimination and retrieval), but could impact on incidental learning (enhancing aided recall, which only requires discrimination).

The question then arises as to whether enduring involvement with a sport has the capacity to modify the potential memory impact on the viewer in regard to aided and unaided recall. In this regard, Park and Hastak (1994) say that levels of involvement do not affect aided recall. It could also be further argued that field sponsorship does not induce unaided recall except for subjects who are highly involved in the sports event. Processing a sports event should logically require less effort for highly involved television viewers. They invest comparatively less effort in processing the game and thus allocate more capacity to processing the field-sponsorship stimuli (Pham, 1992). These viewers process brand names faster than less involved viewers (d'Ydewalle et al., 1988), owing to their expertise or to their domain knowledge (Celsi & Olson, 1988) as discussed previously.

In the light of the previous discussion, the following hypotheses are proposed.

H2a: Field sponsorship impacts on aided recall (of the sponsor's name), whatever the level of enduring involvement of television viewers.

H2b: The effectiveness of field sponsorship depends on the level of

enduring involvement of television viewers as far as unaided recall is concerned.

Enduring Involvement: The Case of Television Sponsorship

Because the salience of television sponsorship stimuli (length, primacy/recency) increases the viewer's opportunity and ability to process information, as compared to field sponsorship stimuli, and because the informative content of these television stimuli is low (i.e., limited to brand name), it is proposed that motivation and ability do not lead to better memory performance (recognition and unaided recall) for involved subjects as compared to uninvolved subjects. Consequently, the following hypotheses are given.

H3a: Television sponsorship impacts on unaided recall (of sponsor names), whatever the level of enduring involvement of television viewers.

H3b: TV sponsorship impacts on aided recall (of sponsor names), whatever the level of enduring involvement of TV viewers.

MEMORY AND THE COMBINATION OF FIELD AND TELEVISION SPONSORSHIP

Edell and Keller (1989) have explained that a stimulus can activate the memorization process by reactivating in the memory a similar message element previously assimilated. It might be expected that a viewer exposed to television sponsorship stimuli before the actual sports broadcast would recognize more easily the sponsor's insignia on billboards during the broadcast. Similarly, after the broadcast, a television sponsorship message should help the viewer remember the sponsor insignia previously seen on field sponsorship displays.

However, given the fact that during broadcasts of sports events, the viewer focuses essentially on the sport activity, and given that attention and processing capacities are limited in such circumstances, it seems unlikely that field sponsorship stimuli could reactivate earlier encoded stimuli (such as television sponsorship stimuli seen immediately before the sports broadcast) in this way. Furthermore the reactivation process could well be interrupted because processing becomes distracted (i.e., by the game) because information does not remain in the memory long enough for higher levels of interaction of the field–television sponsorship stimuli to occur. This argument is supported in the literature, as it has been established that a competing distractor (e.g., the sports event) can inhibit cognitive activity.

In such situations, to increase the likelihood of viewers linking various stimuli, the sponsor should aim (a) to enhance encoding opportu-

nities and (b) reduce the processing time required (MacInnis, Moorman, & Jaworski, 1991). Each of these situations will be considered in turn.

Enhancing Encoding Opportunity—The Combination of Field and Television Stimuli

In terms of effects on memory, repetition of stimuli that are similar but not identical (e.g., field- vs. television-sponsorship stimuli) is more effective than repetition of totally identical signals (Schumann, Petty, & Clemons, 1990). Current literature links this theory of encoding variability to the multiple pathways that form in the memory after exposure to different versions of a stimulus. These pathways act as retrieval cues that improve recall. Thus, multiplying the formats of sponsor stimuli (field vs. television) should increase viewers' attention and provide the opportunities to link these stimuli together. This would imply that, in terms of impact, the combination of field and television sponsorship is more effective than television sponsorship or field sponsorship alone.

Although this rationale is attractive, it does not seem to apply to sponsorship communication, because of the very different characteristics of field and television sponsorship stimuli and the differing pathways through which they are processed. To increase the probability that viewers process the link between television and field sponsorship, it is necessary to ease perception of this link or to shorten the time necessary to process it. These abilities seem to go together with enduring involvement for sports; that is, only those viewers who have a high level of enduring involvement are likely to perceive the message similarities and experience the memory effect.

Reducing the Processing Time

A sports event broadcast is a competitive, dense-with-messages environment, with multiple stimuli from multiple brands, especially when television and field sponsorship are combined. Because of the low-involvement nature of the sponsorship communication context (similar to that described by Krugman, 1972), allocating resources to processing television sponsorship may inhibit efforts to retrieve field sponsorship stimuli.

For a viewer to link a television sponsor to a field sponsor, it is first necessary to sort, from all field-sponsor messages seen during the broadcast (for instance, there were 12 different field sponsors associated with the 1998 Soccer World Cup), the particular message that is placed by the television sponsor.

Nonetheless, linking field and television sponsorship could be done, but only by highly involved television viewers, because they: (a) behave like experts and try to sustain/develop their domain knowledge (Okechuku, 1992) by gathering information (e.g., knowing everything about

sponsors of a sport); (b) have a more structured knowledge of brands usually associated with sports; and thus (c) process brand names faster and with less effort than uninvolved people (d'Ydewalle et al., 1988).

In the case of viewers with low enduring involvement operating in this competitive message-dense environment, the various stimuli may be processed with less ability and mental elaboration, resulting in more retrieval interference (Burke & Srull, 1988) and more inhibited retrieval. Accordingly, recall of field and television sponsorship stimuli in combination could be lower than recall of "television-only" sponsorship and perhaps of "field-only" sponsorship.

Therefore, it is assumed that television sponsorship, field sponsorship, and enduring involvement interact. Furthermore, because of the high level of processing involved, this interaction comes into play only in the case of unaided recall. It is recognized that involvement is unlikely to affect memory effectiveness at the mere recognition (aided recall) level.

Based on these observations, the following hypothesis is given.

H4a: There is an interaction between field sponsorship, television sponsorship, and enduring involvement in the case of unaided recall.

H4b: There is an interaction between field sponsorship and television sponsorship in the case of unaided recall, for *highly* involved television viewers.

At the level of aided recall, the salience of television sponsorship stimuli is such that maximum recognition is reached by this means alone with no possible additional impact arising from exposure to field sponsorship.

Therefore, H4 has a third part.

H4c: There is negative interaction between field sponsorship and television sponsorship in the case of aided recall, whatever the level of enduring involvement of television viewers.

METHODOLOGY

In order to test the hypotheses proposed above, an experimental research study was designed and implemented. This is now reported with a commentary on the results. Important conclusions are identified.

Experimentation Process

The subjects were 240 young adults (mean age 18.3 years, 56% female) assigned to a $2 \times 2 \times 2$ factorial design (Table 1). Subjects were divided

Table 1. Intergroup Complete Factorial Design

Enduring Involvement	Television Sponsorship			
	Absent		Present	
	Field Sponsorship		Field Sponsorship	
	Absent	Present	Absent	Present
High	A1 (control)	B1	C1	D1
Low	A2 (control)	B2	C2	D2

into eight groups of 30 with the use of a blocking technique, based on a previous measurement of their enduring involvement in televised basketball games. The experiment was presented as research to evaluate the factors of interest of sports broadcasts (cover story).

A week later, data gathering took place in a laboratory setting. Each participant was alone in front of a television monitor. A 20-min video clip devised to look like a sports program was broadcast, followed by a first questionnaire to clear short-term memories and to check the credibility of the cover story (the analysis confirmed this credibility). Immediately following the first questionnaire, a second questionnaire was administered that contained the critical dependent measures: unaided and aided recall.

Manipulation of Sponsorship Stimuli

Three types of sport sponsorship stimuli were studied: Television sponsorship only, field sponsorship only, and the combination of the two. Each group of subjects was exposed to only one of four simulated sports reports (one control [A] and three experimental [B, C, and D] reports—Table 1) edited by Canal + (French television network) for the purpose of the experiment.

These programs were identical in terms of sports scenarios but varied with respect to the presence/absence of sponsor stimuli (Table 1).

Each video report consisted of three sections:

1. A general item about soccer was used to divert respondents and diminish potential primacy and recency effects (3'30").
2. The second section was the target broadcast—the basketball report (12'30"), which was a filmed summary of an official first division basketball game.
3. This was a talk about soccer (3'30").

The “soccer reports” (also handled by Canal +) were used to better

disguise intent with respondents and strengthen their perception of actually viewing an excerpt from a real sports show on Canal +.

It was also necessary to use control groups who were shown the basketball report without any sponsorship presence whatsoever. Retrieval of sponsor brand names depends on accessing, from memory, the recording of the association between the sport event and the sponsor (Crimmins & Horn, 1996). Viewers, trying to remember which brand sponsored a sport event, may capitalize on their general knowledge about a sport or about a brand without requiring recollection of the learning episode (Quester, 1997; Pham & Johar, 1997). Thus, it is essential to subtract from experimental results the effect of previous exposure and inference.

Two target sponsors were the focus of the experiment. In the case of field sponsorship, 10 additional sponsors were shown (to simulate the realistic competitive field sponsor environment). In the case of television sponsorship, the 2 target sponsors appear together and for the same duration (4.7 s) immediately before and after the basketball game.

Measurement of the Involvement Construct

Enduring involvement for basketball broadcasts was evaluated with the use of the items relating to enduring dimension of the Laurent and Kapferer (1985) Involvement Profile Scale (I.P.S.), with the use of a 5-point scale anchored by “fully in agreement” and by “not at all in agreement.”

Measurement of Dependent Variables: Memorization

In order to rate effectiveness on learning and memory, “recall” and “recognition” are commonly used dependent variables (Singh, Rothschild, & Churchill, 1988). Each respondent was given an unaided recall test (“top of mind” type), followed by an aided-recall test. Eighteen brands were presented (two target sponsors plus 16 bogus/dummy or distracting stimuli). Unaided and aided recall were coded as follows: 0 = no target sponsor mentioned, 1 = 1 target sponsor mentioned, and 2 = 2 target sponsors mentioned.

RESULTS

The results of the study are now presented with the use of the two dependent variables, unaided and aided recall, as section headings.

The statistical analysis relies essentially on classical *t* tests and variance analysis (ANOVA); *t* tests are frequently used to check if means are statistically different (e.g., mean computed on the recall scores of those exposed to field sponsorship vs. mean computed on the recall

scores of those exposed to TV sponsorship). Therefore these t tests are especially suited to our hypotheses H1a and H1b, which are related to whether there is any difference between two population means. The t values are directly reported in the text along with their corresponding probability (reflecting the level of significance of the difference between the two means; the lower the better, to the extent that this probability refers to the null hypothesis of “no difference”). ANOVA is implemented when one aims to detect main effect (e.g., the influence of enduring involvement on aided or unaided scores for all the subjects, as in H3a and H3b) or when one looks at the potential existence of interaction between explanatory variables (as in H2a, H2b, H4a, H4b, and H4c). The ANOVA results are reported in Tables 2–4.

Unaided Recall

The results of the experiment show that both field sponsorship and television sponsorship are effective in terms of unaided recall. However, the responses suggest that field sponsorship seems capable of leaving only a marginal effect (Figure 1 and Table 2 [see the η^2 value]). As hypothesized (H1a), television sponsorship appears to be significantly more effective than field sponsorship in its impact on unaided recall ($t(118) = 1.21$; $p = .034$).

The detailed results as mapped in Figure 1 show that unaided recall of sponsor stimuli is higher for subjects exposed to television sponsorship than for all others, that is, both those exposed to field sponsorship only and those in the control group ($M_{TV2} = 0.17$ vs. $M_{Field} = 0.08$ and $M_{Ctrl} = 0.02$). Table 2 presents the details of the variance analysis. It is evident that television sponsorship accounts for only 4.6% of unaided-recall variance (Table 2, Television $\eta^2 = 0.046$).

In the case of field sponsorship, the effect of exposure on the unaided recall of viewers might be described as marginal [$F(1;232) = 3.04$; $p = .082$ (Figure 1, Table 2)], and does not provide clear evidence (the p value being only .082) to suggest that such exposure on its own is capable of instigating significant variance. Field sponsorship accounts indeed for only 1.3% of unaided recall variance (Table 2, Field $\eta^2 = 0.013$).

A significant effect arising from exposure to field sponsorship is in fact only present among subjects with high enduring involvement (H2b) as is clearly evident from the data [$F(1,116) = 5.50$; $p = .020$ (Figure 2, Table 3)]. Managers should thus note that to be effective in pursuing the objective of unaided recall, field sponsorship has to be seen by television viewers who are characterized by high levels of enduring involvement in sporting events.

On the contrary, the effects of TV sponsorship on unaided recall [$F(1,232) = 11.06$; $p = .001$ (Figure 1, Table 2)] are not influenced by enduring involvement. Therefore, there is no interaction between TV

Table 2. Analysis of Variance for Unaided Recall.

Source of Variation	SS ^a	df ^b	MS ^c	F ^d	Sig of F ^e	η^{2f}
Within + Residual	38.43	232	0.17			
Television	1.84	1	1.84	11.09	0.001	0.046
Field	0.50	1	0.50	3.04	0.082	0.013
Endurinvol	0.10	1	0.10	0.63	0.429	0.03
Television by field	0.04	1	0.04	0.23	0.635	0.01
Television by endurinvol ^h	0.10	1	0.10	0.63	0.429	0.03
Field by endurinvol ^h	0.50	1	0.50	3.04	0.082	0.013
Television by field by endurinvol ^h	0.94	1	0.94	5.66	0.018	0.024
(Model)	4.03	7	0.58	3.47	0.001	
(Total)	42.46	239	0.18			
R-squared ^g	0.095					

^aSums of squares (SS): Differences between scores and their means. The many varieties of analysis of variance are conveniently summarized in terms of SS. The power of results is increased by increasing the intergroups SS (e.g., TV = 1.84) or decreasing the intragroup SS (i.e., within groups = 38.43).

^bDegrees of freedom: Number associated with a test statistic that is used in determining the observed significance level. The intragroups *df* (within groups = 232) is the number of scores (240) – the number of groups (8; cf. Table 1). The intergroups *df* (or model *df*) is the number of groups (8) – 1. Total *df* (239) is the sum of the intragroup *df* (232) and the intergroup *df* (7).

^cThe division of SS by *df* produces the mean square (MS). The intragroups MS (within groups = 0.17) and the intergroups MS (e.g., TV = 1.84) provide the scores needed to compute an *F*.

^dOnce *F* is computed, it is tested against the critical *F* obtained from a critical values of the *F* distribution table. The numerator is the intergroup *DF* (e.g., TV = 1) and the denominator is the intragroup *DF* (within groups = 232) at desired alpha level (e.g., 5%) of significance. If computed *F* exceeds critical *F*, the hypothesis that there is a difference among the means in the groups is accepted.

^eThe observed significance level of *F*. If this probability is small enough (usually <0.05 or 0.01), the hypothesis that there is a relationship between the dependent (e.g., recall) and independent variables (e.g., presence of TV sponsorship) is not rejected. If this probability is <0.05, there is an effect (e.g., effect of TV sponsorship).

^fEta squared (η^2) can be interpreted as the proportion of variance in the dependent variable explained by variation in the independent variable. The greater the value of η^2 , the greater the effect of a variable (max = 1, min = 0).

^gR squared is the proportion of the variation in the dependent variable explained by the model. It can range in value from 0 to 1. Small values indicate that the model does not fit the data well.

^hEndurinvol = Enduring involvement.

Table 3. Analysis of Variance for Unaided Recall of Involved Subjects.

Source of Variation	SS	df	MS	F	Sig of F	η^2
Within + Residual	20.90	116	0.18			
Television	1.41	1	1.41	7.82	0.006	0.063
Field	1.01	1	1.01	5.50	0.020	0.046
Television by field	0.68	1	0.68	3.75	0.055	0.031
Model	3.09	3	1.03	5.72	0.001	
Total	23.99	119	0.20			
R-squared	0.129					

Table 4. Analysis of Variance for Aided Recall.

Source of Variation	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	Sig of <i>F</i>	η^2
Within + Residual	115.5	232	0.50			
Television	28.70	1	28.70	57.69	0.000	0.199
Field	5.70	1	5.70	11.46	0.001	0.047
Endurinvol ^a	3.04	1	3.04	6.10	0.014	0.026
Television by field	6.34	1	6.34	12.74	0.000	0.052
Television by endurinvol ^a	0.94	1	0.94	1.88	0.171	0.008
Field by endurinvol ^a	0.04	1	0.04	0.08	0.784	0.000
Television by field by endurinvol ^a	0.10	1	0.10	0.21	0.648	0.001
(Model)	44.86	7	6.41	12.88	0.000	
(Total)	160.3	239	0.67			
<i>R</i> -squared	0.280					

^aEnduring involvement.

sponsorship and enduring involvement ($TV \times$ Enduring involvement $F < 1$; see Table 2). H3a is thus supported by the data emerging from the research.

As hypothesized (H4a), field and TV sponsorship do not interact positively, except in the case of highly involved viewers. The contrast between the results achieved in the low-involvement situation and the high-involvement situation is striking (Figure 3, $M_{L.Inv} = 0.14$ vs. $M_{H.Inv} = 0.44$). Among involved television viewers, the “TV $\times$ field” condition does in fact result in higher unaided recall ratings than does the “television-only” condition. ($M = 0.44$ vs. $M = 0.1$).

From a manager’s perspective it would seem inappropriate to invest

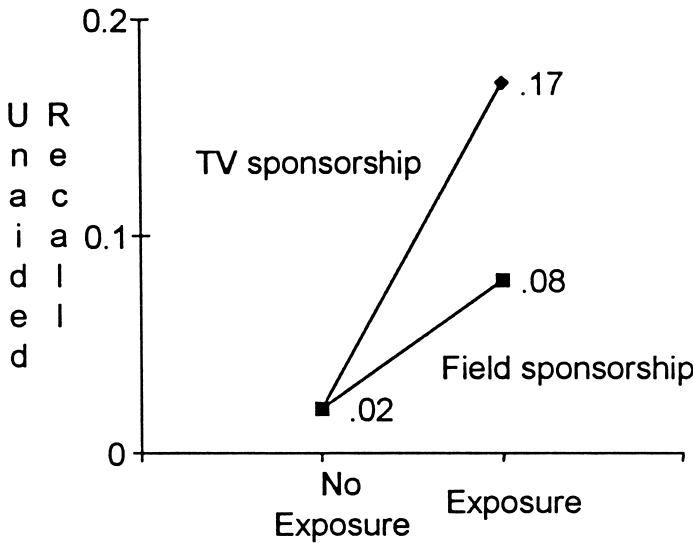


Figure 1. Main effects of field and television sponsorship on unaided recall.

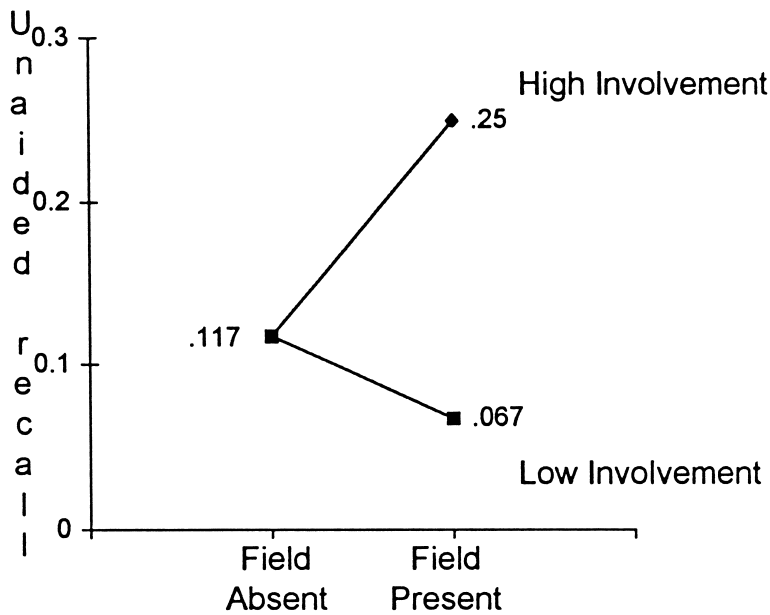


Figure 2. Field sponsorship and enduring involvement interaction on unaided recall.

in a combination of field and television sponsorship in pursuit of unaided recall objectives, if involved people are not being specifically targeted.

The importance and potency of “television” sponsorship stimuli are borne out by the fact that such stimuli seem able to explain more variance of unaided recall than “field” sponsorship stimuli on their own or the combination of “television $\times$ field $\times$ involvement” (see Tables 2 and 3).

Therefore, managers must be aware that the variation in unaided recall scores explained by a sponsorship campaign, which includes television and field stimuli, would not exceed 13% ($R^2 = 0.129$; see Table 3 for the details relating to this option) for involved subjects. Therefore, even in the case of an involved audience the bulk of memorization is driven by factors other than exposure to field- and TV-sponsorship campaigns. These other factors might include the nature of the product/service, the saliency of the brand, the historical links between the brand and sponsorship, or the intensity of the affective reactions that are stimulated.

Aided Recall

Logically, because recognition/aided recall is more subject to learning (as previously discussed, see Singh et al., 1988), it would be expected that significant effects of television- and field-sponsorship stimuli as well as the effect of enduring involvement would be evident from the analysis of the data. It is clear from Figure 4 (and the data presented

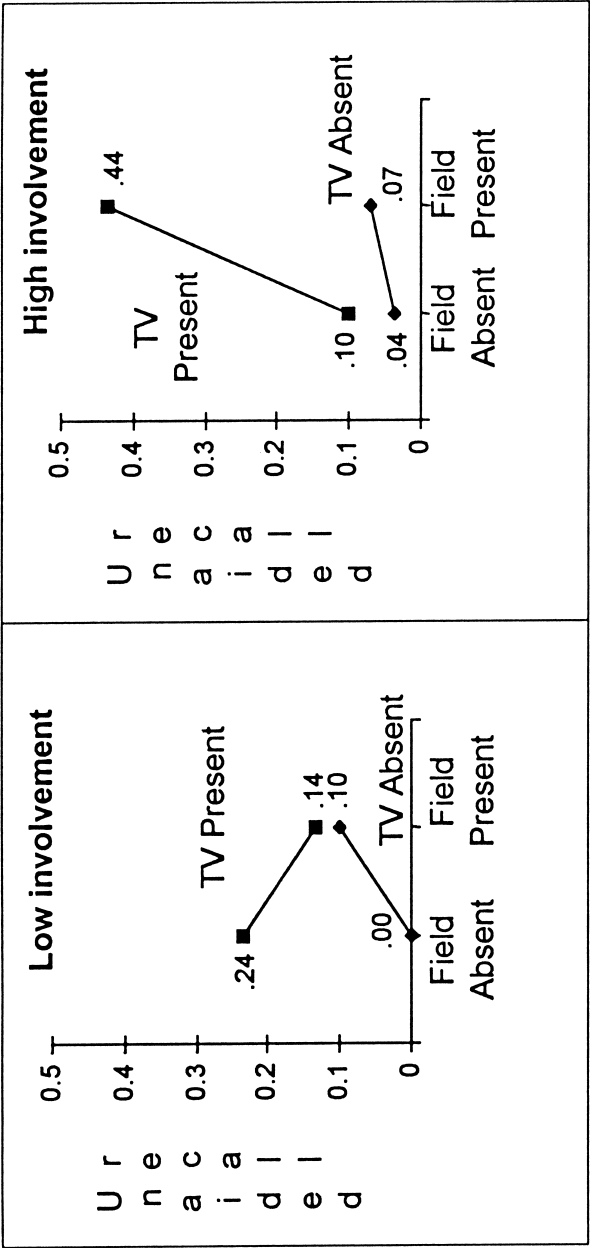


Figure 3. Field × television × enduring involvement interaction on unaided recall.

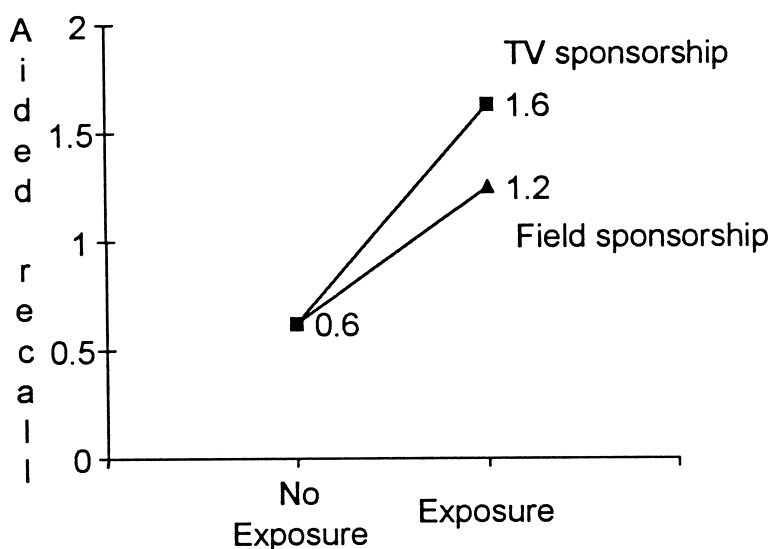


Figure 4. Main effects of field and television sponsorship on aided recall.

in Table 4) that enduring involvement and both field sponsorship and television sponsorship are effective in terms of aided recall.

Once again it appears that television-sponsorship stimuli are more effective than field-sponsorship stimuli. Aided recall of sponsors is measurably better in the case of subjects exposed to television sponsorship than in the case of those exposed to field sponsorship ($M_{TV} = 1.63$ vs. $M_{Field} = 1.25$; $t(128) = 2.92$, $p = .004$). The impact of field sponsorship is four times lower compared to the variance accounted for by television sponsorship (Table 4: $\eta^2 = 0.199$ vs. $\eta^2 = 0.047$).

It had been hypothesised that television sponsorship stimuli influence aided recall whatever the subjects level of involvement (H3b), and the results prove this within the scope of this research [$F(1,232) = 57.69$; $p < .001$ (Table 4)]. Furthermore, TV sponsorship does not interact with enduring involvement [$F(1,232) = 1.88$; $p = .171$ (Table 4)]. Therefore, H3b is accepted.

The effect of enduring involvement on recognition is not dependent on the nature of the sponsorship stimuli, namely, television stimuli vs. field stimuli. However, because of its marginal impact (it accounts for $<3\%$ of sponsor recognition—Table 4: $\eta^2 = 0.026$), managers will not need to pay attention to this factor. Many communication managers invest in a combination of television and field sponsorship and they expect a multiplicative effect. The data would seem to indicate that the interaction of television-sponsorship stimuli and field-sponsorship stimuli does feature, but it is a negative interaction for viewers with low enduring involvement.

As was anticipated, there is clear evidence of a ceiling effect. Figure

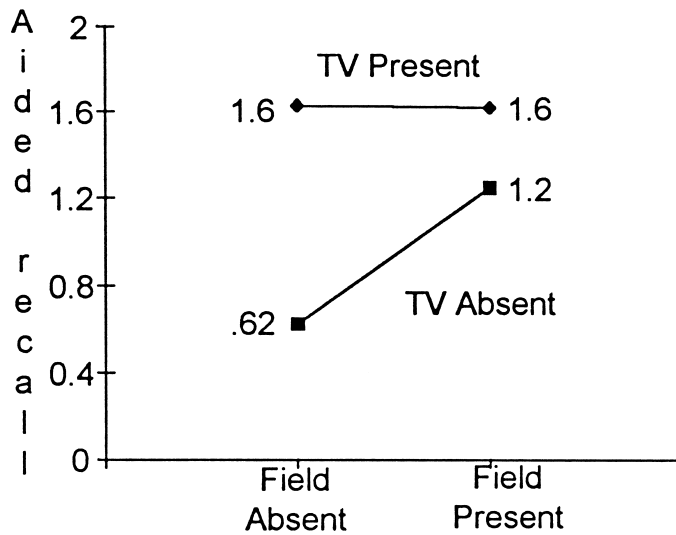


Figure 5. Field- and television-sponsorship interaction on aided recall.

5 shows that exposure to television-sponsorship stimuli alone has a significant impact on aided recall. It is also evident that “field $\times$ television” produces significantly higher results than the “field-only” condition [$t(118) = 2.83; p = .005$]. However, the combination of television stimuli and field stimuli produces no additional variance whatsoever over that achieved by “television-only,” which implies that there is a ceiling on recognition effect and that this can be achieved by exposure to the television-sponsorship stimuli alone.

In effect the “field $\times$ television” condition does not imply a better recognition than the “television-only” condition [$M_{\text{Field} \times \text{TV}} = 1.62$ vs. $M_{\text{TV}} = 1.63; t(118) = 0.14; p = .89$]. There are important implications for marketing managers. If the primary objective is aided recall/recognition, investing in field sponsorship when one is already investing in television sponsorship adds no value whatsoever.

DISCUSSION AND IMPLICATIONS

The study sought to test a range of hypotheses through the experimental design outlined previously. We summarize all our results in Table 5. A number of issues of particular interest are briefly discussed next.

Television vs. Field

It is obvious from the analysis of the data that in comparison to the control group, both field-sponsorship stimuli and television-sponsorship stimuli are effective in terms of memorization. However, field-sponsor-

Table 5. Summary of Tested Hypotheses.

Hypothesis	Result	Sig of <i>F</i> (or <i>t</i>)
H1a	Supported	$t(118) = 1.21; p = .034$
H1b	Supported	$t(118) = 2.92; p = .004$
H2a	Supported	$F(1,232) = 11.46; p < .001$
H2b	Supported (with a 10% error margin)	$F(1,232) = 3.04; p = .082$
H3a	Supported	$F_{\text{unaided}}(1,232) = 11.09; p = .001$
H3b	Supported	$F_{\text{aided}}(1,232) = 57.69; p < .001$
H4a	Supported	$F(1,232) = 5.66; p = .018$
H4b	Supported (with a 10% error margin)	$F(1,116) = 3.75; p = .055$
H4c	Supported	$F(1,232) = 12.74; p \leq .001$

ship stimuli seem capable of leaving only a superficial memory trace, illustrated in this study through recognition tests that rely merely on discrimination. Television-sponsorship stimuli, in contrast, are shown to have influence on both unaided recall and recognition. For subjects exposed to television-sponsorship stimuli, the variance accounted for is always higher than in the case of those exposed to field-sponsorship stimuli and this holds true in the case of both aided recall/recognition and unaided recall.

The analysis suggests that television sponsorship accounts for 3.5 (in the case of unaided recall) to 4 times (in the case of recognition) more variance than field-sponsorship stimuli, although shown for only one-tenth of the time duration (2×4.7 s in the case of television sponsorship vs. a total of 89 s [average for the two target sponsors] aggregated across multiple appearances, in the case of field sponsorship). These findings contradict the belief implicit in the practice of sponsors who frequently calculate their return on investment by translating their seconds of message visibility into an advertising budget, based on the cost of conventional media advertising for a similar length of time. A growing number of sponsors now question this media-equivalence approach to the evaluation of sponsorship effectiveness, and these findings provide them with support for their concerns.

For some managers, the relative cost of the two sponsorship modalities precludes them from the television-sponsorship option, and the question arises as to whether the memorization performance of field-sponsorship stimuli, employed on a solus basis, might be improved. A manager seeking to provide viewers with increased opportunity for processing field-sponsorship stimuli has a number of options. The key parameters that can be varied are the length of the sequence during which sponsors are visible, and the intensity of distraction associated with the sport. The choice of sport can determine the scope of these parameters. For example, a major sponsor of a Formula One team will enjoy longer sequences of visibility than a sponsor whose logo is displayed around the perimeter of a soccer field. In relation to the distraction issue, a

racing car draws attention to itself, and hence to the sponsor's logo, which may well cover the entire visible surface of the car (e.g., Benetton and Ford Racing teams). In the case of soccer, by way of contrast, attention is drawn to the sport actions, rather than to the sponsors' billboards.

There are, however, a number of tactics available now that can help resolve these problems by increasing exposure time. The number of sponsor messages displayed on "pitch-side" perimeter boards could be reduced, or rotating boards, which display a single logo/message around the whole of the field at any one time, can also be used.

Involved vs. Uninvolved

As was emphasized in the presentation of recognition results, an enduring involvement main effect can be identified in the data in relation to Table 4 [$F(1,232) = 6.10; p = .014$]. It is clear that the more involved the viewer, the higher the recognition scores. This prompts a reconsideration of the relationship between information processing *during* exposure to sponsorship and retrieval of information that was encoded *prior* to such exposure. Significantly, in the case of recognition, there is no interaction between enduring involvement and the various modalities of sponsorship examined in this research, suggesting that the effect of enduring involvement on recognition does not depend at all on the types of sponsorship stimuli.

Overall, the analysis suggests that enduring involvement is a minor factor in terms of the effectiveness of sponsorship. Involvement may, however, play a crucial role in promoting the effectiveness of sponsorship, because it is a central factor in bringing the individual to watch sports events, more frequently and for longer periods of time, and thus is instrumental in achieving and extending exposure to sponsors' messages.

The analysis also identifies a gap that remains to be bridged between involvement with a sports event *per se* and involvement with the identified sponsors of this event. Enduring involvement with a sport neither obviates nor compensates for the low-involvement nature of communication through the sponsorship modalities examined in this study, at least from the perspective of viewer's motivation. The authors are of the view that it is low motivation to process information rather than the absence of ability to process that is responsible for this low impact. It could be argued that if attention is drawn and curiosity is raised, motivation to process is likely to increase. New technologies (particularly those that permit the embedding of virtual images) will make it possible to manipulate the logos used in field-sponsorship displays, thus making them animated and intriguing, in other words more noticeable. It may thus be possible to develop strong logical and creatively attractive links between field-sponsorship stimuli and television-sponsorship stimuli for

the television viewer, which in turn could increase the motivation to process information.

Aided vs. Unaided Recall

As far as the R^2 are concerned, let us stress that they are very low for unaided recall (0.095 for all the subjects and 0.129 for the involved subjects) and satisfactory for aided recall (0.28 for all the subjects and 0.34 for the involved ones). This is consistent with existing research findings “. . . that recognition responds better to learning than does unaided recall” (Singh, Rothschild, & Churchill, 1988).

The question remains as to which memorization tasks are best accomplished through exposing viewers to sponsorship stimuli and which modalities of sponsorship are likely to be of use in achieving specific objectives.

Even though television-sponsorship stimuli influence both recognition and unaided recall, the impact on recognition is significantly greater. As has been demonstrated in this study, the variance accounted for by television sponsorship is four times higher in terms of recognition than in terms of unaided recall.

This would seem to suggest that sponsorship, regardless of the modality chosen, is essentially concerned with influencing memorization at a superficial level, and thus is more suited to recognition-related objectives. Therefore, it could be extrapolated that communication through sponsorship seems more suited to products associated with low involvement.

However, this capacity of sponsorship stimuli to leave a memory trace, even if only at a superficial level, does offer a major advantage in that it could enhance access to the fundamental meaning of the brand (Pham & Vanhuele, 1997). Association in a consumer's mind between a supported event and its sponsors is one of the justifications of sponsorship activity of the kind under study.

LIMITATIONS

Although it can be difficult to create low-involvement conditions in a laboratory setting, the research benefited both from the natural distraction induced by the sport action and the credibility of the sports broadcast provided by Canal +. It is important, however, to draw attention to the limitations associated with the research undertaken.

One of the major limitations of this experiment is that it did not test the effects of the length of the experimental sports report as an influencing variable. Perhaps with longer exposure, the impact of field sponsorship would be stronger.

In order to avoid the risk of contamination effects, a single-exposure

approach was adopted in the research study. However, it is known that sponsorship is a communication method that works best over time, and it is important that future research in this field devises alternative methodologies that address the impact of repeated exposure to sponsorship stimuli.

CONCLUSIONS

The objective of this research was to isolate and measure the impact on memory of “field sponsorship only,” “television sponsorship only,” and of the combination of the two, against the background of the level of enduring involvement with the sport of the “subjects/television viewers.” This was the first time that the impact of the “television sponsorship $\times$ field sponsorship” interaction had been tested. The main results of this research are as follows.

- TV sponsorship is the most effective modality both in the cases of aided and unaided recall.
- Field sponsorship has a significant and positive impact on aided recall only.
- There is a positive interaction between TV and field sponsorship on unaided recall when the audience is involved but let us stress that this effect is marginal.
- There is a negative interaction between TV and field sponsorship on aided recall whatever the level of enduring involvement. Thus, the additional investment in field sponsorship in pursuit of this specific objective delivers no added value.

It might further be suggested, contrary to the general trend of argument in the literature, that the combination of the two modalities (television $\times$ field), employed in the pursuit of both recognition and unaided recall outcomes, does not represent the recommended strategy of communication, as seems to be implicitly assumed in many sponsorship campaigns. This study definitively shows that the percentage of variance that the combination accounts for is relatively minor. Even within the acknowledged limits of the present study there are grounds for concluding that sponsorship is not well suited to the building of unaided recall even with highly involved subjects. However, the situation may be very different if obvious or historical links already exist between the brands (e.g., Adidas, Nike, and Coca-Cola) and the domain (sports events).

By way of contrast for brands seeking to improve their unaided recall scores, sponsorship seems to emerge as among the most effective communication options.

The study confirms that the television sponsorship option remains the cornerstone of communication through sponsorship, for both recognition and unaided recall objectives. These results support, in the context of sponsorship communication, the primacy of the “opportunity to process” over the “motivation and ability to process” in achieving outcomes.

Further research is urgently needed to examine the positive and negative interactions between television and field sponsorship that might prove a valuable contribution to both academic and applied research. This is especially true because this area is currently experiencing important changes (e.g., for the Soccer 2002 World Cup, ISL is trying to reserve television sponsorship to official events sponsors and, in so doing, to prevent ambush sponsorship).

This initial research into the relationships between various forms of sponsorship stimuli has provided insights into the effectiveness of the respective modes but represents only a first tentative step towards a larger understanding.

REFERENCES

- Burke, R. R., & Srull, T. K. (1988). Competitive interference and consumer memory for advertising. *Journal of Consumer Research*, 15, 55–68.
- Celsi, R. L., & Olson, J. C. (1988). The role of involvement in attention and comprehension processes. *Journal of Consumer Research*, 15, 210–224.
- Crimmins, J., & Horn, M. (1996). Sponsorship: From management ego trip to marketing success. *Journal of Advertising Research*, 36, 11–21.
- Derbaix, C., Gérard, P., & Lardinoit, T. (1994). Essai de conceptualisation d'une activité éminemment pratique: Le parrainage. *Recherche et Applications en Marketing*, 2, 43–67.
- Droits television: Le dernier Mondial libre d'accès pour les téléspectateurs? (1998, June 16). *Les Echos*.
- d'Ydewalle, G., Vanden Abeele, P., Van Rensberger, J., & Coucke, P. (1988). Incidental processing of advertisement while watching soccer-games broadcast. In M. Gruneberg, P. Morris, & R. Sykes (Eds.), *Practical aspects of memory: Current research and issues* (pp. 478–483). New York: John Wiley & Sons.
- Edell, J. A., & Keller, K. L. (1989). The information processing of coordinated media campaigns. *Journal of Marketing Research*, 2, 149–163.
- European Group of Television Advertising. (1993). *European guide to television sponsorship*, Brussels.
- FIFA: Un business qui rapporte gros. (1998, January 29). *Le Nouvel Economiste*.
- Keller, K. L. (1991). Memory and evaluation effects in competitive advertising environments. *Journal of Consumer Research*, 17, 463–476.
- Krugman, H. E. (1972). Why three exposures may be enough. *Journal of Advertising Research*, 12, 11–14.

- Krugman, H. E. (1985). Point of view: Measuring memory an industry dilemma. *Journal of Advertising Research*, 25, 49–51.
- Laurent, G., & Kapferer, J. N. (1985). Measuring consumer involvement profiles. *Journal of Marketing Research*, 22, 41–53.
- MacInnis, D. J., Moorman, C., & Jaworski, B. J. (1991). Enhancing and measuring consumers' motivation, opportunity, and ability to process brand information from ads. *Journal of Marketing*, 55, 32–53.
- Meenaghan, T. (1991). The role of sponsorship in the marketing communications mix. *International Journal of Advertising*, 10, 35–47.
- Meenaghan, T. (1994). Ambush marketing: Immoral or imaginative practice. *Journal of Advertising Research*, 34, 77–88.
- Meenaghan, T. (1998). Ambush marketing: Corporate strategy and consumer reaction. *Psychology & Marketing*, 15, 305–322.
- Okechuku, C. (1992). The relationship of prior knowledge and involvement to advertising recall and evaluation (pp. 115–132). Elsevier Science Publishers.
- Park, C. W., & Hastak, M. (1994). Memory-based product judgements: Effects of involvement at encoding and retrieval. *Journal of Consumer Research*, 21, 534–547.
- Petty, R. E., Wells, F. L., & Brock, T. C. (1976). Distraction can enhance or reduce yielding to propaganda: Thought disruption vs. effort justification. *Journal of Personality and Social Psychology*, 34, 874–884.
- Pham, M. T. (1992). Effect of involvement, arousal and pleasure on the recognition of sponsorship stimuli. *Advances in Consumer Research*, 19, 1–9.
- Pham, M. T., & Johar, G. V. (1997). Contingent processes of source identification. *Journal of Consumer Research*, 24, 249–265.
- Pham, M. T., & Vanhuele, M. (1997). Analyzing the memory impact of advertising fragments. *Marketing Letters*, 8, 407–417.
- Pieters, R. G. M., & Bijmolt, T. H. A. (1997). Consumer memory for television advertising: A field study of duration, serial position, and competition effects. *Journal of Consumer Research*, 23, 362–372.
- Quester, P. G. (1997). Awareness as measure of sponsorship effectiveness: The Adelaide Formula One Grand Prix and evidence of incidental ambush effects. *Journal of Marketing Communication*, 3, 1–20.
- Richins, M. L., & Bloch, P. H. (1986). After the new wears off: The temporal context of product involvement. *Journal of Consumer Research*, 13, 281–285.
- Schumann, D. W., Petty, R. E., & Clemons, D. S. (1990). Predicting the effectiveness of different strategies of advertising variation: A test of the repetition-variation hypotheses. *Journal of Consumer Research*, 3, 192–202.
- Singh, S. N., Rothschild, M. L., & Churchill, G. A., Jr. (1988). Recognition versus recall as measures of television commercial forgetting. *Journal of Marketing Research*, 3, 72–80.
- Sponsoring: Les bons comptes de la coupe du monde. (1998, July 7). *La Tribune*.
- Tavassoli, N. T., Shultz, C. J., & Fitzsimons, G. J. (1995). Program involvement: Are moderate levels best for ad memory and attitude toward the ad? *Journal of Advertising Research*, 35, 61–72.
- Townley, S., Harrington, D., & Couchman, N. (1998). The legal and practical prevention of ambush marketing in sports. *Psychology & Marketing*, 15, 333–348.

- Young, D. R., & Bellezza, F. S. (1982). Encoding variability, memory organization, and repetition effect. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 8, 545–559.
- Zaichkowsky, J. L. (1985). Measuring the involvement construct. *Journal of Consumer Research*, 12, 341–352.

The authors are indebted to Ph. Gerard (UCL, Belgium) for his invaluable assistance and to the guest editor, A. Bemmaor (ESSEC, Paris) and R. Darmon (ESSEC, Paris) for insightful comments on earlier drafts of the article.

Correspondence regarding this article should be sent to: Christian Derbaix, Marketing Department, F.U.C.A.M, Chaussée de Binche, 151 B-7000 Mons. Belgium (derbaix@message.fucam.ac.be).